Huawei Access Node Hardware/Environmental Fault Resolution SOP (MALL-ACC-03)

Node: MALL-ACC-03 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8656

Description

The specified failure domain of Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss) is a critical issue that can severely impact the operability of the Huawei Access Node MALL-ACC-03 at the Malleswaram Node site. This scenario is particularly relevant due to the intrinsic dependencies between the Huawei Access Node's hardware components, specifically the line cards and power supplies. Line cards are responsible for forwarding data packets between the node and the external network, whereas power supplies ensure uninterrupted operation of the node by providing redundant power feeding. ASIC (Application-Specific Integrated Circuit) memory parity errors may occur due to hardware malfunctions, overheating, or transient failures, while line card crashes can be attributed to overloading, software bugs, or environmental stressors. Power supply redundancy loss may result from a faulty or malfunctioning power supply unit (PSU) or a failure in the redundant PSU configuration. Given the distributed nature of the network and the reliance on Huawei's CLI dialect, remote hands will be dispatched to the Malleswaram Node to facilitate troubleshooting and resolution of the issue. The primary objective is to identify the root cause of the Hardware/Environmental Fault and rectify it through targeted commands, thus restoring normal operation of the Huawei Access Node.

Pre-requisite Checks

- Dispatch remote hands to the Malleswaram Node to facilitate troubleshooting and resolution of the issue.

- Verify the status of redundant links to ensure uninterrupted data transmission.

- Check the power supply configuration to confirm the presence of redundant power feeding.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware malfunctions, overheating, or transient failures.	High
Line card crashes attributed to overloading, software bugs, or environmental stressors.	Medium
Power supply redundancy loss resulting from a faulty or malfunctioning power supply unit (PSU) or a failure in the redu	Low

Diagnostic Commands & Expected Outputs

```
Command: display power-supply, Expected Output: Power supply
information, including status, voltage, and current., Description:
This command isolates the power supply configuration and status,
facilitating identification of potential issues related to power
supply redundancy loss.
```

```
Command: display line-card, Expected Output: Line card information, including status, temperature, and fan speed., Description: This command isolates the line card configuration and status, enabling detection of potential issues related to line card crashes or ASIC memory parity errors.

Command: display system redundancy, Expected Output: Redundancy information, including status and configuration., Description: This command isolates the redundancy configuration and status, facilitating identification of potential issues related to power supply redundancy loss or line card crashes.
```

Resolution Steps

Step 1: Step: Verify power supply redundancy and rectify if necessary., Command: replace power-supply , Description: This step ensures that the power supply redundancy is restored by replacing the faulty or malfunctioning power supply unit (PSU).

Step 2: Step: Replace line card if necessary., Command: replace line-card , Description: This step rectifies the issue by replacing the faulty or malfunctioning line card, thus restoring normal operation of the Huawei Access Node.

Step 3: Step: Reset ASIC memory parity errors., Command: reset asic , Description: This step resolves the issue by resetting the ASIC memory parity errors, thus restoring normal operation of the Huawei Access Node.

Post-Check Verification

```
Command: display power-supply, Expected Output: Power supply information, including status, voltage, and current., Description: This command verifies the power supply configuration and status, ensuring that the resolution steps have been successful.

Command: display line-card, Expected Output: Line card information, including status, temperature, and fan speed., Description: This command verifies the line card configuration and status, ensuring that the resolution steps have been successful.

Command: display system redundancy, Expected Output: Redundancy information, including status and configuration., Description: This command verifies the redundancy configuration and status, ensuring that the resolution steps have been successful.
```

Roll-back Procedures

- Step: Restore original power supply configuration., Command: undo replace power-supply , Description: This step safely rolls back the resolution by restoring the original power supply configuration in case the fix fails.

- Step: Restore original line card configuration., Command: undo replace line-card , Description: This step safely rolls back the resolution by restoring the original line card configuration in case the fix fails.

- Step: Restore original ASIC configuration., Command: undo reset asic , Description: This step safely rolls back the resolution by restoring the original ASIC configuration in case the fix fails.